DATABASE MANAGEMENT SYSTEM (ENCS205)

Lab Report Submitted to

K. R. Mangalam University

for

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in
Computer Science and Engineering
Submitted by

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Experiment No: 1

Aim:

To design an Entity Relationship (ER) diagram for the College database and convert the ER diagram into relational tables.

Description:

The system consists of entities such as **Student, Class, Semester, Subject, and Marks**. A student attends a class, enrolls in subjects, and scores marks in each subject. Each class belongs to a specific semester, and a semester can have multiple classes. The marks entity stores internal and final assessment details of students.This ER diagram helps in clearly defining relationships between entities and avoids data redundancy. It provides a strong foundation for designing a relational database for managing student academic records.

Entities and Attributes

1. **STUDENT**
   Represents a student studying in the institution.

   Attributes:

   - **USN (Primary Key)**
   - **SName**
   - **Address**
   - **Phone**

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2. CLASS

Represents a class allotted to students.

Attributes:

- USN
- SSID (Section / Class ID)

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3. SEMESTER

Represents the academic semester.

Attributes:

- SSN
- Sem
- Sec

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4. SUBJECT

Represents subjects taught in the course.

Attributes:

- Subcode (Primary Key)
- Title
- Credits

- **Sem**

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5. **MARKS**
   Represents the marks scored by students.

   Attributes:

   - **USN**
   - **SSD**
   - **Test1**
   - **Test2**
   - **Final**
   - **SCode**

Relationships:

1. STUDENT – CLASS

Relationship: *Attends*

- A student attends a class.
- A class can have many students.

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2. CLASS – SEMESTER

Relationship: *Belongs To*

- A class belongs to one semester.
- A semester can have multiple classes.

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3. STUDENT – SUBJECT

Relationship: *Enrolls*

- A student can enroll in many subjects.
- A subject can have many students.

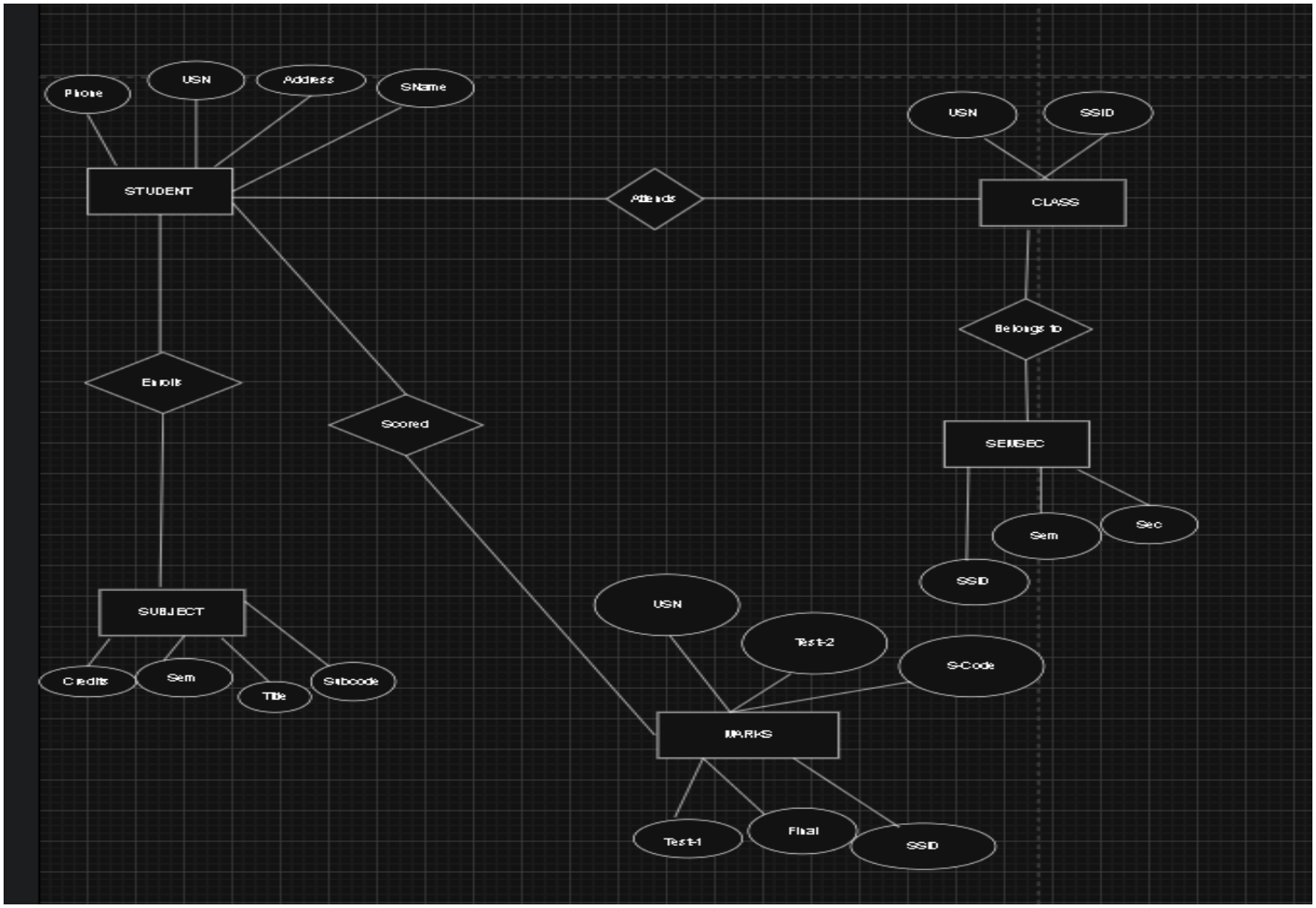
---

4. STUDENT – MARKS

Relationship: *Scored*

- A student scores marks in different subjects.
- Marks depend on the student and subject.

ER diagram:



Conclusion: This ER diagram clearly explains how students are related to classes, semesters, subjects, and marks. It helps in designing a proper database structure for managing academic records efficiently.

Experiment No.2

Aim:

To design an Entity Relationship (ER) diagram for the Company database and convert it into relational tables.

Description:

The system includes entities such as **Employee, Department, Project, and Location**. Employees work in departments, departments control projects, and employees may work on multiple projects. Departments can also be located in different locations. The ER diagram helps in organizing company data, reducing redundancy, and ensuring data integrity.

Entities and Attributes

EMPLOYEE

Represents an employee working in the organization.

Attributes:

- SSN (Primary Key)
- Name
- ▪ Address
- Salary
- Sex
- SuperSSN
- DNo

DEPARTMENT

Represents a department in the organization.

Attributes:

- DNo (Primary Key)
- DName
- MgrSSN
- MgrStartDate

PROJECT

Represents a project handled by the department.

Attributes:

- PNo (Primary Key)
- PName
- PLocation
- DNo

DLOCATION

Represents the location of a department.

Attributes:

- DLoc
- DNo

Relationships:

1. EMPLOYEE – DEPARTMENT
Relationship: *Worked*
An employee works in one department, and a department can have many employees.

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2. DEPARTMENT – PROJECT
Relationship: *Works On*
A department controls many projects, and each project is associated with one department.

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3. EMPLOYEE – PROJECT
Relationship: *Works On*
An employee can work on multiple projects, and a project can have multiple employees. The relationship stores the attribute **Hours**.

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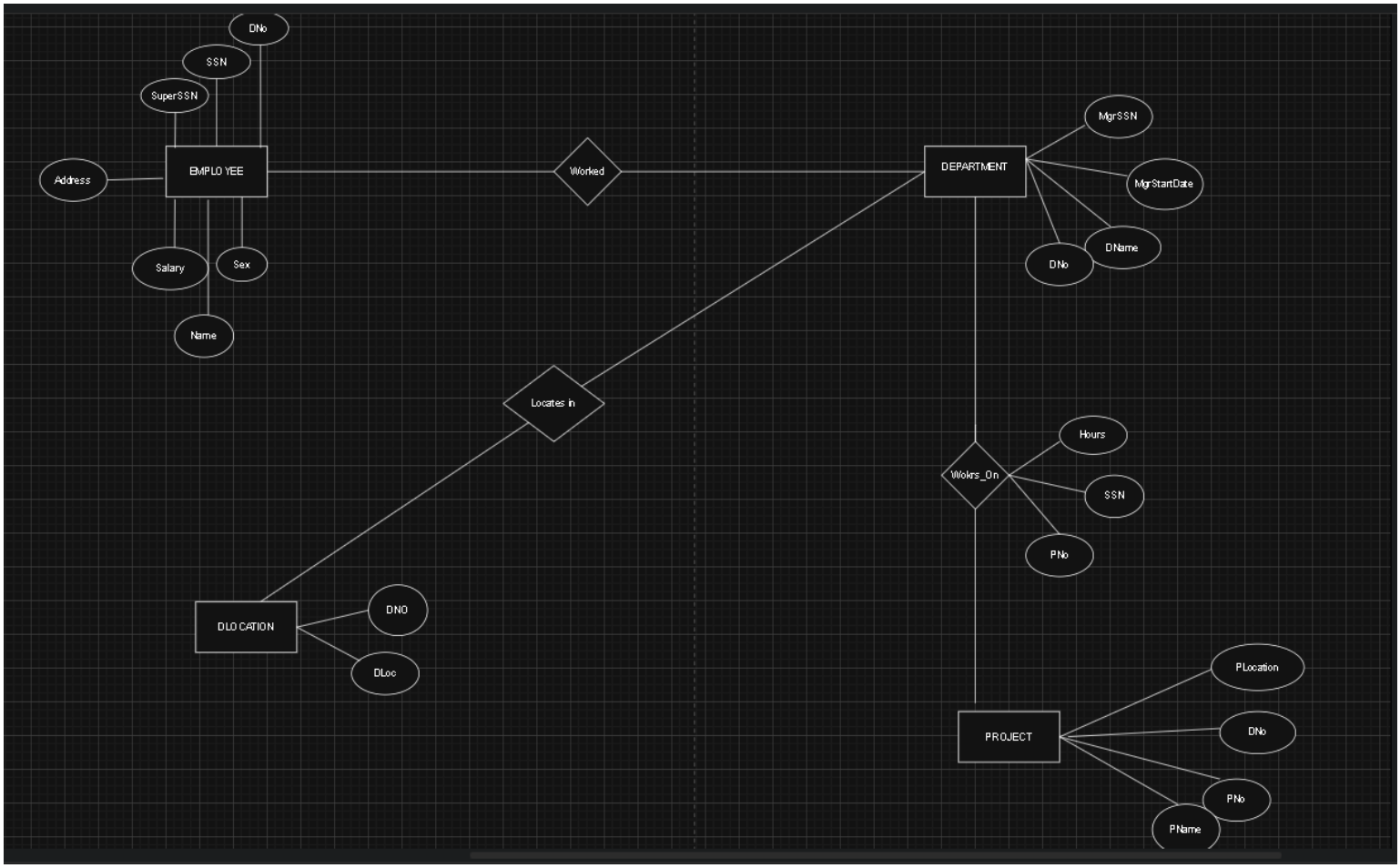
4. DEPARTMENT – LOCATION
Relationship: *Located In*
A department can be located in multiple locations.

ER diagram:



Conclusion:

This ER diagram clearly illustrates the relationships between employees, departments, projects, and locations in a company. It serves as a strong foundation for designing a well-structured and efficient company database system.